

Model Analysis Report

PRS Analytical Pricing Model

MKM-PR-001

Report Date: 27 March 2026

Governance Framework: SR 11-7 / SS1/23 Model Risk Management

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1 Sensitivity Analysis

This section presents the sensitivity of model outputs to key input parameters. Parameter perturbation ranges are chosen to cover plausible operating conditions and stress scenarios as required under SR 11-7 guidance.

Table 1: PRS spread sensitivity to annual hazard rate ($T = 5\text{yr}$, $R = 0$, $r = 0.03$).

λ_{annual}	λ (continuous)	Spread (bp)
0.0010	0.0010	10.0
0.0050	0.0050	50.3
0.0100	0.0100	101.0
0.0200	0.0202	203.3
0.0500	0.0513	518.2
0.1000	0.1054	1,071.6

Table 2: PRS spread sensitivity to risk-free rate ($\lambda_{annual} = 0.01$, $R = 0$, $T = 5\text{yr}$).

r	Spread (bp)
0.0000	100.6
0.0100	100.8
0.0300	101.0
0.0500	101.3
0.1000	101.9

Table 3: PRS spread sensitivity to tenor ($\lambda_{annual} = 0.01$, $R = 0$, $r = 0.03$).

Tenor (yr)	Spread (bp)
1	101.0
2	101.0
3	101.0
5	101.0
7	101.0
10	101.0
20	101.0
30	101.0

2 Stress Testing

This section documents stress testing scenarios applied to the model under extreme but plausible conditions.

2.1 Stress Scenarios

Table 4: Stress testing scenarios for PRS Analytical Pricing Model.

Scenario	Parameter Shock	Severity	Status
Baseline	No shock	—	Passed
Moderate stress	+2 σ inputs	Medium	Pending
Severe stress	+3 σ inputs	High	Pending
Reverse stress	Output threshold breach	Critical	Pending

Detailed stress test results will be populated as scenarios are executed. Refer to the model's validation plan for the full stress testing programme.

3 Model Performance

This section tracks key performance metrics for the model across validation and production environments.

Table 5: Performance metrics for PRS Analytical Pricing Model.

Metric	Value	Status
Unit test pass rate	See Test Results	—
Execution time (single run)	TBD	—
Numerical stability	TBD	—
Reproducibility	Deterministic	OK

This report is produced automatically; human review is required before formal model approval. Supporting artefacts are available in the model documentation directory.